Supplementary Tables for "CovProp: Propagating Upstream Forecast Uncertainty of Future Covariates into Conformal Prediction Intervals"

This artifact-appendix is generated mechanically from the shipped raw results under `results/` — no number is transcribed by hand. Rebuild: `python supplementary/build_supplementary.py --pdf`

1 Robustness ablations

Table 1: Biased upstream-error-log ablation (full grid). "f" is the retained smallest-$|\eta|$ fraction of the error log; "noCal" is the propagation-only variant ($\lambda \equiv 1$).

stream	f	MC cov	MC worst-tercile	MC width	noCal cov
NP	1.0	0.892	0.843	1.770	0.818
NP	0.5	0.900	0.851	1.935	0.800
NP	0.25	0.901	0.852	1.969	0.798
NP	0.1	0.904	0.854	2.001	0.798
FR	1.0	0.875	0.769	1.808	0.755
FR	0.5	0.873	0.767	1.851	0.738
FR	0.25	0.874	0.769	1.855	0.737
FR	0.1	0.873	0.767	1.867	0.735
ETTh1	1.0	0.926	0.918	1.128	0.842
ETTh1	0.5	0.926	0.918	1.147	0.839
ETTh1	0.25	0.926	0.918	1.151	0.838
ETTh1	0.1	0.927	0.919	1.166	0.835
ETTh2	1.0	0.902	0.893	1.819	0.887
ETTh2	0.5	0.904	0.894	1.800	0.889
ETTh2	0.25	0.904	0.894	1.803	0.889
ETTh2	0.1	0.903	0.894	1.783	0.890
ETTm2	1.0	0.915	0.901	3.885	0.877
ETTm2	0.5	0.915	0.902	3.646	0.876
ETTm2	0.25	0.914	0.897	3.759	0.876
ETTm2	0.1	0.916	0.900	3.233	0.877
electricity	1.0	0.885	0.869	2.634	0.918
electricity	0.5	0.887	0.866	2.773	0.897
electricity	0.25	0.889	0.863	2.815	0.893
electricity	0.1	0.891	0.864	2.855	0.890
weather	1.0	0.800	0.774	0.294	0.609
weather	0.5	0.803	0.775	0.335	0.530
weather	0.25	0.807	0.781	0.342	0.513
weather	0.1	0.812	0.786	0.362	0.495

Table 2: Error-pool-size sensitivity, full grid: marginal coverage of CovProp-MC as the upstream-error pool is subsampled to m; the last column is CovProp-Share at the full pool.

stream	m=5	10	25	50	100	200	full	Share(full)
NP	0.898	0.897	0.895	0.893	0.893	0.893	0.892	0.881
FR	0.870	0.874	0.875	0.875	0.876	0.875	0.875	0.876
ETTh1	0.926	0.926	0.927	0.926	0.927	0.926	0.926	0.928
ETTh2	0.904	0.904	0.904	0.902	0.903	0.904	0.902	0.905
ETTm2	0.915	0.915	0.914	0.915	0.914	0.915	0.915	0.917
electricity	0.888	0.886	0.883	0.883	0.883	0.885	0.885	0.880
weather	0.810	0.805	0.802	0.801	0.801	0.801	0.800	0.809

Table 3: Per-instance propagated-width diagnostics, seed 0 (n=1500 windows per stream). $\rho_s(\text{half}, |\eta|)$ is the weak correlation reported in the paper; $\rho_s(\text{half}, \rho_t)$ shows the width does track the label-free share statistic.

stream	$\rho_s(\text{half}, \lvert\eta\rvert)$	$\rho_s(\text{half}, \rho_t)$	$\bar{\rho}_t$	std(half)
NP	-0.117	-0.285	-0.126	0.545
FR	-0.187	-0.177	-0.061	0.854
ETTh1	0.110	0.150	-0.364	0.171
ETTh2	0.137	0.332	-0.029	0.405
ETTm2	-0.065	0.225	-0.109	10.264
electricity	-0.109	-0.340	0.160	0.530
weather	-0.281	-0.746	-0.234	0.036

2 Per-instance and applicability-domain diagnostics

3 Negative result and deployment metrics

Backtest selector (paper §VII-E, Table VIII). Across the 63 preregistered operating points the label-free backtest selector attains mean Winkler 2.157 vs. oracle 2.151 (mean regret 0.24%, median 0.00%, max 2.63%; choose-rate agreement 74.6%). Wilcoxon vs. always-recalibrate: +6.6% ($p = 2.48e-07$); vs. always-share: +13.1% ($p = 8.88e{-}04$). Cross-check vs. the frozen 3-seed run: max relative deviation of seed-averaged backtest means 3.1%.

Table 4: Applicability-domain statistics per stream (`ad_analysis.json`). Skill is upstream forecast skill vs. persistence; share is the width-based propagation share; gain is the Winkler improvement of CovProp-Share over recalibration; spread is the residual coverage spread across share bins.

stream	skill	rel-err	share	gain	bin-spread
NP	0.622	0.340	0.122	0.024	0.007
FR	0.465	0.230	0.267	-0.005	0.008
ETTh1	0.648	0.451	0.098	-0.033	0.006
weather	0.680	0.489	0.153	0.172	0.117
ETTh2	0.993	0.225	0.102	-0.111	0.006
ETTm2	1.031	0.176	0.558	-1.135	0.014
electricity	0.497	0.317	0.017	0.069	0.021

Table 5: Winkler score per stream for the two preregistered hybrid designs vs. the lean roster (lower is better). The pick beats CQR on only 2/7 streams against the preregistered $\geq 5/7$ criterion; adopted as a negative result.

stream	recal	cqr	C-mc	C-share	H-mc_cqr	H-cqr_covwidth
NP	2.722	2.592	2.675	2.655	2.818	2.606
FR	2.515	2.535	2.519	2.526	2.940	2.538
ETTh1	1.337	1.018	1.371	1.381	0.989	1.018
ETTh2	2.000	1.353	2.226	2.221	1.376	1.354
ETTm2	1.969	0.874	4.052	4.203	0.897	1.036
electricity	3.457	2.712	3.310	3.220	2.755	2.702
weather	0.543	0.212	0.459	0.450	0.211	0.213

Table 6: Per-window latency (ms) on 1500 test windows, seed 0. ×recal is the K=8 latency relative to plain recalibration.

stream	recal	MC K=8	K=16	K=32	×recal
NP	0.016	0.044	0.068	0.126	0.244
2.7×					
FR	0.017	0.042	0.068	0.131	0.235
2.4×					
ETTh1	0.018	0.044	0.076	0.135	0.262
2.5×					
ETTh2	0.016	0.045	0.070	0.133	0.245
2.8×					
ETTm2	0.016	0.046	0.072	0.130	0.251
2.9×					
electricity	0.016	0.045	0.074	0.130	0.243
2.8×					
weather	0.018	0.043	0.070	0.125	0.250
2.4×					